STATISTICS ASSIGNMENT

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1. Variables Encountered in Daily Apps

We use many apps such as Instagram, Zomato, and Flipkart in our daily life. These apps contain different types of variables.

No.	Variable	Example	Classification
1	Number of Likes	1,250 likes	Numerical
2	Restaurant Name	Domino's	Categorical
3	Product Price	₹999	Numerical
4	Payment Method	UPI	Categorical
5	Number of Followers	5,000 followers	Numerical
6	Product Category	Electronics	Categorical
7	Delivery Time	35 minutes	Numerical
8	City Name	Ahmedabad	Categorical

Numerical variables are represented by numbers and can generally be measured or counted. **Categorical variables** represent names, groups, or categories.

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2. Classification of Given Variables

Variable	Classification	Reason
Age of a user	Numerical	Age is measured in numbers and can be compared or calculated.
City name	Categorical	City names represent different groups or categories.

Payment status (Paid/Unpaid)	Categorical	Paid and Unpaid are categories rather than numerical measurements.
Product rating	Numerical	A rating such as 1, 2, 3, 4, or 5 is represented using numbers.
Delivery time in minutes	Numerical	Time is measured numerically in minutes.
Cuisine type	Categorical	Indian, Chinese, Italian, etc. represent different categories.

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3. Use of Statistics in a Cricket/IPL App

Feature 1: Player Runs

Statistics can be used to calculate a player's **average runs, total runs, and consistency** across matches. This helps users understand and compare player performance.

Feature 2: Team Name

Statistics can be used to study a team's **number of wins, losses, and winning percentage**. This helps users understand the team's overall performance and compare it with other teams.

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4. Variables Used as Features in a Machine Learning Model

A food delivery app can collect different types of information about users. Some of these variables can be useful for understanding and predicting user behavior.

No.	Variable	Used as a Feature?	Reason
1	Previous Orders	Yes	Shows what food or restaurants the user has ordered from before.
2	Favorite Cuisine	Yes	Helps identify the user's food preferences.
3	Average Order Value	Yes	Helps understand the user's usual spending pattern.

4	Order Frequency	Yes	Shows how often the user places food orders.
5	Delivery Location	Yes	Helps identify nearby restaurants and understand the user's ordering area.

Conclusion

Variables are an important part of statistics and data analysis. They help us collect, organize, and understand information. **Numerical variables** contain measurable values, while **categorical variables** contain different groups or categories.

Apps such as Zomato, Flipkart, Instagram, and cricket apps use these types of data to understand users, analyze performance, and provide better recommendations and services.